PRODUCT REQUIREMENTS DOCUMENT | OWNER: PM (KYC) | FOR: BACKEND, INTEGRATIONS, REVIEW OPS

KYC Verification Waterfall v2 - PRD

The waterfall that finishes running. 7.93% of applicants are not verified against a 5% bar, and half of that is applicants the waterfall simply stopped processing. Thirteen requirements, three of them invariants, and a capacity-gated rollout.

> **This document is self-contained.** It describes the system as it exists today, what is wrong with it, what we are building, and how we ship it. No prior reading is required. Metric definitions and instrumentation live in the companion **ARD / Monitoring Plan**; staffing and sequencing live in the companion **Project Plan**. Every number here is measured from the production KYC record set (2,515,155 applications, enrolled Jan 2023 – May 2026, 107 internal test accounts excluded).

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1. Summary

Every user of our US consumer product must pass identity verification (KYC) before we can open their account. Verification runs as a **waterfall**: a primary identity provider, then one or more secondary providers, then a human reviewer. Today **7.93% of applicants end up not verified**, against a 5% regulatory-and-commercial bar — 199,421 people, of whom roughly 73,663 are over budget.

The dominant cause is not that people fail checks. It is that the waterfall stops running. 101,162 applicants (4.02% of the book, **50.7% of all non-verifications**) failed the primary check and then had **no second check of any kind —** no secondary provider, no human review. Three of them verified. Applicants who failed the same primary check and *were* routed onward recovered at **48.09%**.

v2 makes "stops running" structurally impossible. Every primary failure is classified into a reason, every reason has a defined path, and no applicant can leave the system without landing in one of exactly three terminal states. We are **not** loosening any check, threshold, or standard to hit the bar; the entire improvement comes from routing.

Expected outcome: 7.93% → ~6.0%, with a committed target of **≤6.3%** and a 5.0% stretch (§3 explains why we commit to 6.3% and not 5.0%). **The cost is human review volume: roughly 2.4× today's case load.** That is the real trade this design makes, and it is what gates the rollout.

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2. The system today

2.1 Providers in use

Tier	Provider	What it checks	Applicant friction
L1	**Idology**	Primary identity: name, DOB, address, SSN. Also carries our sanctions/PEP screening signal.	None — invisible
L2 (SSN path)	**LexisNexis**	Identity + SSN verification. Integrated **twice** (`ProviderA_Lexis_Nexis` , `ProviderB_Lexis_Nexis`), both writing to one result field.	None
L2 (SSN path)	**Persona — SSN mode**	SSN verification, documented as the route for "older/legacy accounts".	None
L2 (non-SSN path)	**ACRO**	Automated secondary identity check (name / DOB / address / email).	None
L3	**Persona — IDV mode**	Document capture + selfie. Highest assurance.	**High** — 60–120s of active user effort

Tier	Provider	What it checks	Applicant friction
L4	**Manual review**	Human analyst; final decision authority, including compliance cases.	None (but adds hours–days of wait)

The field `overall_kyc_status` is the sole record of the final outcome.

2.2 The waterfall as documented

1. **L1 Idology.** Pass ⇒ verified, exit. Fail ⇒ log the reason (SSN error vs. non-SSN error) and route on.
2. **L2/L3 split by failure type.**
   - *SSN issue* → LexisNexis (standard accounts) or legacy Persona-SSN (older accounts). **Crucial rule:** if that check surfaces a secondary *non-SSN* issue, move the applicant to the non-SSN track.
   - *Non-SSN issue* (name / DOB / address / email) → ACRO or Persona IDV.
3. **L4 Manual review.** If every automated check fails, a human assesses the case.

2.3 What the production data actually shows

85.67% of applicants follow the documented design. The remaining 14.33% is where the loss is concentrated. Five defects, each measured:

#	Defect	Volume	Verified	Why it matters
D1	**Stranded** — Idology FAIL, then nothing at all ran	**101,162** (4.02%)	**0.003%**	Half of all non-verification. Comparable routed applicants recover at 48.09%.
D2	**Primary bypassed** — Idology never ran; a LexisNexis integration acted as an undocumented second front door	**94,183** (3.74%)	99.08%	A control gap: ~3.7% of the book was approved without our primary identity check.
D3	**PASS did not exit** — clean Idology PASS, yet further checks ran anyway	**131,538** (5.23%)	99.81%	Mostly wasted spend, but **247 applicants passed the primary check and still ended not verified.**
D4	**Safety net missed** — automation exhausted, no human ever looked	**48,598** non-verified	0%	Manual review reached only **30.2%** of the 213,048 applicants who exhausted automation.
D5	**Sanctions signal is invisible** — sanctions/PEP rides on the same undifferentiated FAIL as an ordinary typo	9,994 flagged cases	79.0% cleared	We cannot tell a watchlist hit from an address mismatch at routing time. And **zero** of the 101,162 stranded applicants have any reviewer record, so their risk profile is unobserved.

2.4 Every non-verification, classified

Rule: *two or more checks ran and all failed* = a genuine decline; *one check, then nothing* = an avoidable process loss.

Bucket	Applicants	% of non-verified	Reading
A — no check ever ran	843	0.4%	Never entered the waterfall
B — exactly one check ran, then stopped	101,325	50.8%	D1
C — 2+ checks ran, never reached a human	48,598	24.4%	D4
D — fully assessed including a human, still declined	48,655	24.4%	Working as intended

75.6% of non-verification (A+B+C) is process failure, not a correct decision. Only bucket D — just under a quarter — is the system doing its job. Bucket D is explicitly **not** a target: we found no way to shrink it without weakening a control.

2.5 Two more measured facts the design leans on

- **The crossover rule works.** 8,605 applicants hit the documented SSN→non-SSN crossover and recover at **55.0%** — but only because the right checks happened to fire, not because anything enforces it.
- **Our best tool is barely used.** Persona IDV verifies **86.3%** of the primary-check failures it is tried on, versus 30.7% for failures it is not. It ran on **0.87%** of them (2,507 cases).

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3. Goals

Baselines are measured over the full record set; targets are measured at 100% rollout. The metric IDs (M1, M2, …) are defined in the companion ARD.

#	Goal	Baseline	Target	Metric
G1	**No applicant exits the waterfall undecided**	101,162 stranded (4.02%)	**0**	M2
G2	Reduce non-verification	7.93%	**≤6.3%** (5.0% stretch)	M1
G3	Every applicant gets the primary check	94,183 (3.74%) bypassed it	**100% L1 coverage**	M3
G4	A clean PASS is final	131,538 (5.23%) continued past PASS; 247 lost	**0 post-PASS checks**	M4
G5	The human backstop actually catches	30.2% of automation-exhausted reached a human	**100%**	M5
G6	Sanctions/PEP is a distinct, fast-routed signal	Folded into a bare FAIL; 0 records on stranded users	**100% routed to compliance, p95 ≤4h**	M7
G7	No control is weakened to buy the rate	—	Guardrails within tolerance	G-a, G-b, G-c

Why G2 commits to 6.3% and not 5.0%. The 7.93% → ~6.0% projection applies the observed 48.09% recovery rate of routed applicants to the stranded population. That is a **comparison-group estimate, not a measured outcome** — the stranded group never had a second check, so their true recoverability is unobserved. The residual gap to 5.0% depends on Persona IDV behaving at volume as it does at 0.87% of volume, and on a risk-based auto-decline layer that does not exist yet. We re-baseline this target on live data at Phase 3 rather than commit engineering to a number we cannot yet defend.

One goal works against G2, and we are shipping it anyway. Retiring the primary-check bypass (G3) forces 94,183 applicants/window who currently verify at 99.08% to clear a real primary check first. Some will fail. G3 is a **control fix with a plausibly negative conversion effect**. It ships behind its own flag and is measured separately (M3b) so the two effects are never confused.

Non-goals

- Changing any verification standard, threshold, or the definition of a PASS.
- Reducing the genuine-decline floor (bucket D, 48,655).

- Adding vendors. We fix the routing of what we already pay for.

4. Scope

In scope

1. **L1.5 reason classifier —** a deterministic rule layer (not a vendor) that tags every primary-check failure with exactly one reason class, plus a separable sanctions/PEP flag and a secondary-issue signal.
2. **Orchestration rewrite —** mandatory L1, unconditional PASS-exit, guaranteed terminal state, retry and timeout handling, one feature flag per behavioural change.
3. **Provider role cleanup —** the two LexisNexis integrations unified into one secondary role (L2a); Persona-SSN (L2b) and Persona IDV (L3) exposed as distinct callable tiers; the LexisNexis primary entry point removed.
4. **Sanctions/PEP split —** independent flag with an immediate compliance route.
5. **The SSN→non-SSN crossover**, as an explicit branch rather than an emergent behaviour.
6. **Manual review —** routing, capacity model, and full reason/path metadata on every case.
7. **Event log, dashboards, and alerts** (specified in the ARD).
8. **One-time migration** of the existing stranded backlog.

Out of scope, and why

Out	Why
Risk-based auto-decline (hard-reject low-risk automated failures without a human)	Needs labelled outcome data that v2 itself will generate. This is the main lever to shrink the manual-review cost — it is a v2.1 fast-follow, deliberately not v1.
Shrinking the genuine-decline floor	No evidence it is addressable without weakening a control.
New vendors	Fix routing first.
Applicant-facing UX rework	Except the Persona IDV hand-off, which v1 must not make worse.

5. The v2 design

5.1 Tier roles

Tier	Check	Role in v2	Change from today
L1	Idology	Mandatory primary identity check for **100%** of applicants. Emits identity result **and** a separate sanctions/PEP flag.	Closes D2. Sanctions signal is no longer folded into PASS/FAIL.
L1.5	Reason classifier *(new, internal)*	Maps every L1 FAIL to exactly one of `SANCTIONS_HIT`, `SSN_MISMATCH`, `IDENTITY_ATTR_MISMATCH`, `NO_DATA`. Also emits the secondary-non-SSN-issue signal that drives the crossover.	New. Drives all routing below.
L2a	LexisNexis (single integration)	SSN-path check for **standard** accounts.	Two integrations unified; no longer usable as a primary.

Tier	Check	Role in v2	Change from today
L2b	Persona — SSN mode	SSN-path check for **legacy** accounts. An *alternative* to L2a, never a chain — each applicant gets exactly one.	Restores the documented standard/legacy split. Gate definition is an open dependency (§8).
L2	ACRO	Secondary identity-attribute check, and the landing point for the crossover.	Same role, now reliably reached.
L3	Persona — IDV mode	Highest-assurance automated fallback. Placed **last** on the non-SSN path because it is the most expensive and the only step with real user friction.	Promoted from 0.87% of failures to the standard second non-SSN fallback.
L4	Manual / compliance review	Two entry points: automation exhausted, or an immediate sanctions/PEP route. Final decision authority.	Reaching L4 becomes a guarantee, not a possibility.

5.2 Routing rules, in priority order

```
                     L1 — IDOLOGY (100%, no exceptions)
                     → identity result + sanctions flag

   sanctions/PEP flag            PASS, no flag              FAIL, no flag
          ▼                           ▼                          ▼
 L4 COMPLIANCE                 VERIFIED                 L1.5 REASON CLASSIFIER
 immediate, skips           flow terminates
 all automation

 PASS → VERIFIED                      SSN_MISMATCH   IDENTITY_ATTR_MISMATCH   NO_DATA
 FAIL → HARD REJECT                        ▼                  ▼                  ▼
                                  legacy account?         L2 ACRO          applicable SSN chk
                                   yes → L2b                                + L2 ACRO
                                   no  → L2a            PASS    FAIL        IN PARALLEL
                                  (exactly one)                 ▼
                                                                          any PASS  both FAIL
                                   PASS     FAIL                          → VERIFIED
                                                          L3 IDV      ◄
                                secondary non-SSN
                                issue surfaced?          PASS    FAIL
                                 yes → non-SSN track            ▼
                                 no  → L4                VERIFIED / L4
```

1. **L1 runs on every applicant.** No bypass, no exception.
2. **Sanctions/PEP flag ⇒ L4 compliance immediately**, regardless of the identity result, skipping all automated fallbacks. Compliance decides; a FAIL there is a hard reject.
3. **Clean PASS with no flag ⇒ VERIFIED, flow terminates.** Nothing downstream may fire. *(Fixes D3.)*
4. **FAIL with no flag ⇒ L1.5 assigns exactly one reason class**, which sets the path:
   - `SSN_MISMATCH` → legacy ? **L2b Persona-SSN** : **L2a LexisNexis** — exactly one, never both. PASS ⇒ verified. FAIL ⇒ **crossover check**: a secondary non-SSN issue surfaced? Yes → non-SSN track (ACRO → IDV → L4). No → L4.
   - `IDENTITY_ATTR_MISMATCH` → **L2 ACRO** → (FAIL) **L3 IDV** → (FAIL) **L4**.
   - `NO_DATA` → applicable SSN check **and** ACRO in parallel → any PASS verifies; both FAIL → **L3 IDV** → (FAIL) **L4**.
5. **L4 is the last stop on every path.** PASS ⇒ verified; FAIL ⇒ hard reject.

5.3 Stop conditions

Terminal state	Reached when	Why it is final
VERIFIED	Any tier returns PASS on the applicant's assigned path, sanctions flag absent or cleared	Symmetric with hard reject — once set, nothing may re-open it
HARD REJECT	(a) compliance confirms a sanctions/PEP match, or (b) every automated check on the assigned path failed **and** a human declined	(a) is a legal decline, never a routing problem. (b) is the genuine-decline floor.
IN MANUAL REVIEW	Automation exhausted, or sanctions flag raised	Transitional — must resolve to VERIFIED or HARD REJECT

Deliberately not a stop condition: an automated FAIL on its own, at any tier. Every path reaches a human before a final no. This *strengthens* our control posture relative to today, where 48,598 applicants were declined by automation with no human ever looking.

6. Functional requirements

Numbered for traceability into tickets and tests. **FR-1, FR-2 and FR-3 are invariants** — assertions enforced in code and monitored continuously, not best-effort behaviours.

ID	Requirement
FR-1	Every application record must, at any point after 72h, be in **exactly one** of `VERIFIED`, `HARD_REJECT`, `IN_MANUAL_REVIEW`. "Failed a check, nothing pending, no decision" must be unrepresentable in the state model.
FR-2	No application may reach a terminal state without an L1 event.
FR-3	Once L1 returns PASS with no sanctions flag, no downstream check — automated or manual — may be dispatched for that application.
FR-4	L1.5 must assign exactly one reason class to every L1 FAIL; unclassifiable failures take `NO_DATA` and its fan-out path.
FR-5	The sanctions/PEP flag is evaluated independently of the identity result and takes routing precedence over every other class.
FR-6	On the SSN path, exactly one of L2a / L2b executes per application.
FR-7	On an SSN-path FAIL, the crossover condition must be evaluated before falling through to L4.
FR-8	Provider error or timeout is **not** a FAIL. The application parks in a retry queue (3 attempts, exponential backoff, 30-minute window) and escalates to L4 if unresolved. No applicant is rejected because a vendor was down.
FR-9	Every check attempt and every routing decision emits an immutable event (schema in the ARD). Auditability must not depend on inferring the path from per-provider result columns.
FR-10	Every L4 case carries the reason class, every provider result, and the full path taken, so the reviewer is not re-deriving what happened.
FR-11	Each behavioural change — routing engine, PASS-exit, bypass retirement, IDV promotion — sits behind an independent feature flag and can be reverted without reverting the others.
FR-12	Cohort assignment is a stable hash of the application id. An application never changes cohort mid-flight.

ID	Requirement
FR-13	Persona IDV must instrument **invitation, start, and completion separately.** An abandoned IDV must be distinguishable from a decline.

7. User flows

7.1 Success — ~84.8% of applicants

Submit → **L1 Idology** → clean PASS, no sanctions flag → **VERIFIED**, flow terminates. Latency unchanged (<1s), no applicant action, no downstream call permitted.

Today 79.57% of applicants take this path cleanly; v2 adds the 5.23% who currently pass and then get extra checks anyway (D3), bringing it to **84.80%**. This path is untouched in substance and is the regression risk we watch hardest during rollout.

7.2 Error / fallback — the fix

L1 FAIL, no sanctions flag → L1.5 classifies → routed per §5.2.

Reason class	Path	Applicant experience
`SSN_MISMATCH`	L2a **or** L2b → crossover check → non-SSN track or L4	Silent
`IDENTITY_ATTR_MISMATCH`	L2 ACRO → L3 IDV → L4	Silent until L3
`NO_DATA`	SSN check + ACRO in parallel → L3 IDV → L4	Silent until L3

Only **L3 Persona IDV** is visible to the applicant: an in-app prompt for a document photo and a selfie, 60–120 seconds of effort. It is deliberately last, so anyone who clears at the cheap, frictionless ACRO step never sees it.

7.3 Escalation

- **Sanctions/PEP flag → L4 compliance queue immediately**, p95 ≤4h to queue, ahead of all automated fallbacks. Sized for a mostly-false-positive load: of 9,994 flagged cases in the baseline, **79.0% cleared**. Compliance decides; FAIL is a hard reject with no appeal path in v1.
- **Automation exhausted → L4 identity review queue**, with full case metadata (FR-10).
- **Provider outage → hold, do not decide** (FR-8).
- **Queue breach —** if L4 depth exceeds the phase's capacity model, the rollout ramp halts automatically at its current percentage (it does **not** auto-revert; see §9).

7.4 Contingency: if the reason classifier cannot be built

The classifier assumes Idology returns structured reason fields. **This is unconfirmed** and is the single load-bearing assumption in the design (§8). If Idology returns only PASS/FAIL, v1 falls back to **reasonless fan-out**: on any FAIL, run the applicable SSN-path check and ACRO in parallel, then L3, then L4 — i.e. the `NO_DATA` path for everyone.

This still delivers G1, G3, G4 and G5. It costs materially more per failed applicant (~2 automated calls instead of ~1.3) and **loses the sanctions fast-path**, which is the real loss. The decision is forced in week 2, before any classifier code is written.

8. Assumptions and open dependencies

#	Assumption	Status	If it fails
A1	Idology's API returns structured reason fields and a separable sanctions/PEP flag	**Unconfirmed — verify first**	Fall back to §7.4
A2	"Legacy account" has an operational definition (a cutover date, product cohort, or explicit flag)	**Undefined.** No signal in our data — age, enrollment vintage and bank partner all fail to reproduce it	Ship v1 routing **all** `SSN_MISMATCH` → L2a, logged as a documented deviation. Add L2b when Account Services can define the gate.
A3	Manual review capacity can grow ~2.4× on the rollout timeline	Gating constraint — see Project Plan	Ramp slows. Capacity, not code, sets the pace.
A4	Real provider unit costs and latency SLAs	Not in our data; all cost figures are illustrative unit costs on **real volumes**	Cost thresholds re-set before Phase 2
A5	Persona IDV completion holds at 10–20× current volume	Unknown — abandonment today is indistinguishable from decline	FR-13 makes it measurable; IDV ships on its own flag

9. Rollout and migration

The ramp is gated by manual-review capacity, not engineering readiness. v2 roughly 2.4×'s review volume (~21,900 → ~51,600 cases/yr; ≈8 additional analyst FTE at illustrative rates). Ramping faster than hiring converts a routing fix into a queue backlog — a worse applicant experience than the stranding we are fixing.

Phase	Traffic	Duration	Entry gate	Exit gate
P0 Shadow	100% mirrored, **0% acting**	2 wks	v2 deployed, dashboards live	Classifier ≥95% agreement with reviewer-labelled ground truth; projected L4 volume within ±20% of model
P1 Canary	5%	2 wks	P0 exit + 2 analysts onboarded	Stranded rate = 0 in cohort; no P1 alerts; L4 turnaround p90 <72h
P2 Ramp	25%	3 wks	+3 analysts	Non-verification in cohort ≤6.5%; IDV completion ≥65%; guardrails flat
P3 Majority	50%	3 wks	+3 analysts; queue turnaround stable	Guardrails within tolerance; **G2 target re-baselined on live data**
P4 Full	100%	—	All gates green	Legacy path deleted after 30 days dark

Risk controls built into the ramp

- **Shadow before acting.** P0 runs v2 on all traffic making zero real decisions — the only phase where a wrong classifier costs nothing. Given A1, skipping it would be the worst decision available.
- **One flag per change** (FR-11). Bypass retirement in particular is *expected* to cost verification rate; bundled with the routing fix it would mask a genuine win.
- **Halt, do not auto-revert.** An unresolved P1 alert freezes the ramp at its current percentage. Rollback is a human decision — flipping routing mid-application is its own failure mode. The legacy path stays warm through P4.

- **A control cohort survives to P4.** The metric that decides whether v2 was right is post-verification fraud rate, and it lags 90 days. Cutting to 100% early destroys the comparison.
- **Sticky randomisation** by application-id hash (FR-12).

Migration

- **Stranded backlog.** The 101,162 existing stranded applicants are a live liability, not history. Re-run them through v2 in batches, oldest first, throttled to **≤20% of daily review capacity**, starting after P2. Re-consent and notification copy owned by Compliance and Legal before batch one. At the 48.09% comparison recovery rate this is the largest single recovery in the programme — and the one most likely to swamp the queue if run unthrottled. Its recoveries are reported separately from the steady-state rate.
- **Data migration: none destructive.** v2 writes a new append-only decision-event log alongside the existing result columns. Legacy columns keep being written through P4, so every dashboard has a continuous series across the cutover, and a daily reconciliation job compares the two.

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10. Glossary

Term	Meaning
Waterfall	The ordered sequence of identity checks: primary → secondary → human
Stranded	An application with a failed check, nothing pending, and no decision — the defect v2 eliminates
Reason class	One of `SANCTIONS_HIT`, `SSN_MISMATCH`, `IDENTITY_ATTR_MISMATCH`, `NO_DATA`
Crossover	The documented rule moving an applicant from the SSN track to the non-SSN track when an SSN-path check surfaces a non-SSN issue
PEP	Politically Exposed Person — a screening category requiring enhanced due diligence
Terminal state	`VERIFIED`, `HARD_REJECT`, or `IN_MANUAL_REVIEW`
Guardrail	A metric proving we did not buy conversion by weakening a control